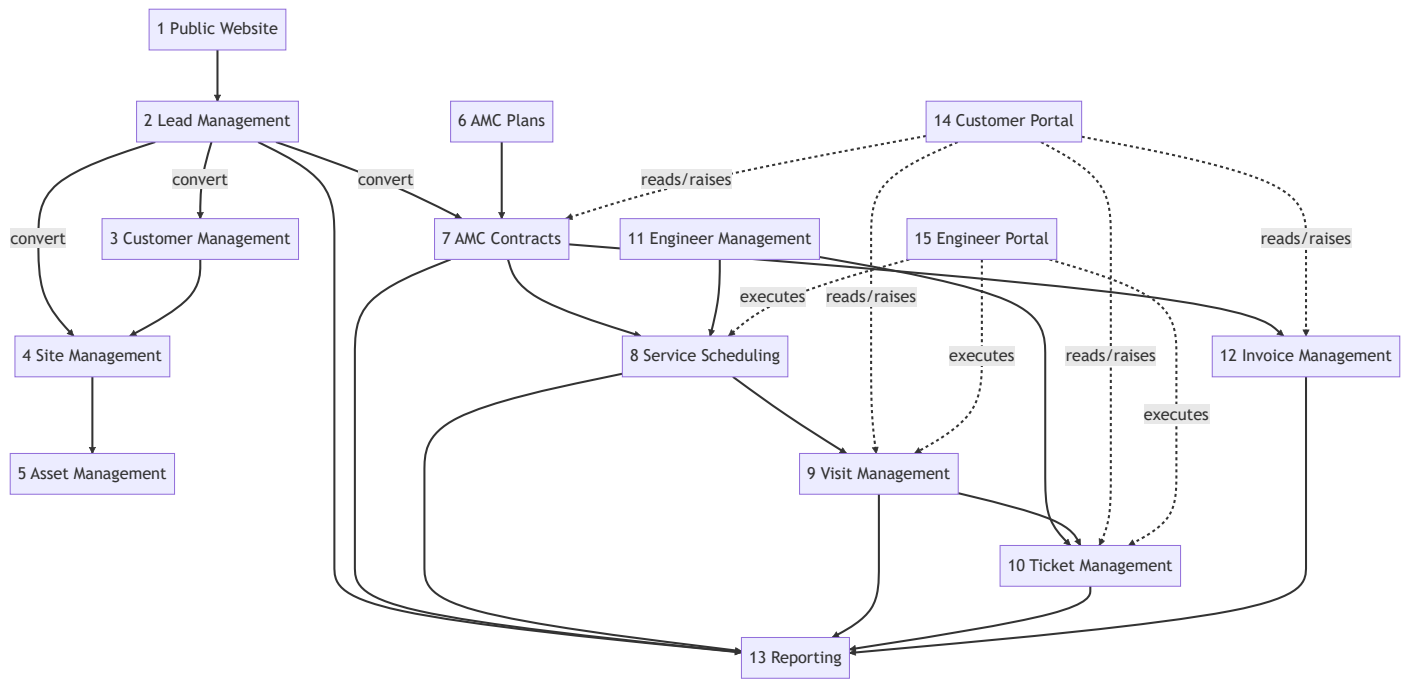


Module Architecture

Project: Aarvii CCTV AMC Management System **Phase:** D0 — Project Foundation Documentation **Source of truth:** requirements-freeze-v1.md §4 (Approved Modules)

Each of the **15 approved modules** is described with responsibilities, boundaries, dependencies, and future extensibility. All are **business modules** on the frozen Ashraak platform — optional, decoupled, replaceable, independently deployable (*business-module-policy*). Dependencies on Core are via published contracts only (freeze §20).



1. Public Website

Aspect	Detail
Responsibilities	Showcase pages (Home, About, Services, AMC Services, Contact, Gallery, Testimonials, Login); Get Quote + AMC Inquiry forms; reuse existing content (§2)
Boundaries	Anonymous only; no business data displayed; submits inquiries to Lead Management
Dependencies	Lead Management (inquiry → lead); platform Auth (login entry)
Future extensibility	New content pages, SEO enhancements, additional inquiry form types

2. Lead Management

Aspect	Detail
Responsibilities	Lead capture (auto from website), pipeline statuses (§10), conversion creating Customer + Site + Initial AMC Contract; Lead Created/Converted notifications (§17)
Boundaries	Owens leads only; conversion <i>initiates</i> records in Customer/Site/AMC modules but doesn't own them; not a general CRM (§21)
Dependencies	Customer/Site/AMC Contracts (conversion), Notifications
Future extensibility	Additional lead sources, scoring, campaign attribution (change request required)

3. Customer Management

Aspect	Detail
Responsibilities	Customer master records; customer profile self-service; customer ↔ user account linkage (§3, §5)
Boundaries	Owens customers, not sites/contracts; credentials owned by platform Auth
Dependencies	Platform Auth/Users (accounts, profile), Site Management
Future extensibility	Customer segmentation, additional profile attributes

4. Site Management

Aspect	Detail
Responsibilities	Sites per customer; ≤3 contact persons per site; site as aggregation point for assets, contract, visits, tickets, invoices (§5, §6)
Boundaries	One site = one customer; enforces single active AMC contract per site (with AMC Contracts)
Dependencies	Customer Management
Future extensibility	Site geo-metadata, site grouping/regions

5. Asset Management

Aspect	Detail
Responsibilities	Per-site summary counts : Camera, DVR, NVR, Hard Disk, Switch, Router, Monitor (+ optional Brand/Model/Remarks) (§7)
Boundaries	No individual device tracking (§7); no inventory/stock (§21)
Dependencies	Site Management
Future extensibility	Per-device tracking would be a V2+ change request — current model deliberately summary-only

6. AMC Plans

Aspect	Detail
Responsibilities	Plan catalog (Silver/Gold/Platinum examples) with Price, Visit Frequency, Included Services, SLA; plan versioning (§9)
Boundaries	Plan changes never mutate historical contracts (§9); plans are admin-managed only (§15)
Dependencies	None on other CCTV modules (referenced by AMC Contracts)
Future extensibility	New plan tiers and versions are native; pricing models per change request

7. AMC Contracts

Aspect	Detail
Responsibilities	Contract Master + Terms model; renewal history; active-term resolution; customer renewal requests; expiry reminders; AMC Contract PDF (§3, §8, §17, §19)
Boundaries	One active contract per site (§6); customer sees active term only, admin sees history (§8); pins plan version at term creation (§9)
Dependencies	Site Management, AMC Plans, Invoice Management (term-linked invoices), Notifications, platform Files (PDF)
Future extensibility	New term types/renewal flows via change request; webhook events for contract lifecycle

8. Service Scheduling

Aspect	Detail
Responsibilities	Auto-generate visits from AMC plan frequency; schedule lifecycle (Planned→...→Cancelled); admin rescheduling; mandatory engineer assignment ; Visit Scheduled notifications (§11, §17)
Boundaries	Owns the schedule; execution evidence belongs to Visit Management
Dependencies	AMC Contracts (frequency source), Engineer Management (assignment), Notifications
Future extensibility	Capacity/route planning per change request

9. Visit Management

Aspect	Detail
Responsibilities	Visit execution: selfie, GPS (lat/long/timestamp), before/during/after photos, videos, signature, remarks; completion checklist; admin approval workflow ; Visit Report PDF; Visit Completed notification (§12, §13, §17, §19)
Boundaries	Customer visibility strictly post-approval (§13); evidence requirements non-bypassable (§12)
Dependencies	Service Scheduling, platform Files (media), Notifications, Ticket Management (tickets raised during visits)

Aspect	Detail
Future extensibility	Additional evidence types or checklists via change request

10. Ticket Management

Aspect	Detail
Responsibilities	Complaint tickets: lifecycle (Open→...→Reopened), priorities (Low→Critical), creation by Customer/Admin/Engineer, customer reopen; ticket notifications (§14, §17)
Boundaries	Not a generic helpdesk; scoped to customer/site service complaints
Dependencies	Customer/Site (context), Engineer Management (assignment), Notifications
Future extensibility	SLA timers tied to plan SLA, escalation rules via change request

11. Engineer Management

Aspect	Detail
Responsibilities	Engineer records and admin management; engineer work-queue surfaces (assigned visits/tickets) (§3, §15)
Boundaries	Engineers cannot manage customers, plans, or contracts (§15); no payroll/attendance/geo-tracking (§21)
Dependencies	Platform Auth/Users (engineer accounts, roles)
Future extensibility	Skills, territories, availability via change request

12. Invoice Management

Aspect	Detail
Responsibilities	Invoice lifecycle (Draft→Generated→Sent→Paid/Cancelled); link to AMC Contract Term ; customer PDF download; Invoice Generated notification (§16, §17, §19)
Boundaries	No accounting features (§16); no payment gateway (§21)
Dependencies	AMC Contracts (terms), Notifications, platform Files (PDF)
Future extensibility	Payment gateway and accounting integrations are explicit V2+ change requests

13. Reporting

Aspect	Detail
Responsibilities	Admin reporting across leads, contracts, visits, tickets, invoices, engineers (§2)
Boundaries	Read-only aggregation; no data ownership

Aspect	Detail
Dependencies	Reads from all business modules
Future extensibility	New reports additive; dashboards/charts per platform chart contract

14. Customer Portal

Aspect	Detail
Responsibilities	Customer-facing experience (web + mobile): dashboard, AMC details, service history, upcoming visits, tickets, invoices, profile, password reset (\$2)
Boundaries	Own-data only (\$3); approved reports only (\$13); active term only (\$8)
Dependencies	AMC Contracts, Visit, Ticket, Invoice modules; platform Auth; Theme Engine (platform-ui)
Future extensibility	New self-service features via change request

15. Engineer Portal

Aspect	Detail
Responsibilities	Engineer-facing experience (web + mobile): assigned work, visit reporting, media capture, signature, ticket creation (\$2)
Boundaries	Assigned work only; \$15 restrictions apply
Dependencies	Scheduling, Visit, Ticket modules; platform Auth; Theme Engine; mobile offline support (\$18)
Future extensibility	Additional field workflows via change request

Cross-cutting platform reuse (freeze \$20)

Platform capability	Consumed by
Authentication / Roles / Permissions	All portals and APIs
Files	Visit media, signatures, PDFs
Notifications (email) + SMS provider	Modules 2, 7, 8, 9, 10, 12; Auth OTPs
Audit	All business actions (observer)
Theme Engine	All web portal UIs
Mobile Foundation	Customer & Engineer apps
API Keys / Webhooks	External integration surfaces

No duplicate implementation of any of the above.

Related documents

- [high-level-design.md](#)
- [application-architecture.md](#)
- [business-rules.md](#)
- Platform module conventions: [../extending/add-backend-module.md](#)